Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 20



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Recap of Previous Lecture



- Topic — Practice questions
- Topic — Finite group
- Topic — Addition modulo 'm' $\oplus_m$
- Topic — Multiplication modulo 'm' $\otimes_m$

{0}, {1}, {1, -1}
'+'
• {Multiplication}

Slide

Topics to be Covered



Topic — Example of groups w.r.t. $\oplus_m$ & $\otimes_m$

Topic — Order of an element of the group (G,*)

Slide



Topic : Finite Group

eg. $\{0\}$ is a finite group of order = '1' w.r.t. binary opⁿ addition.

eg. $\{1\}$ is a finite group of order = '1' w.r.t. binary opⁿ multiplication

eg. $\{1, -1\}$ is a finite group of order = '2' w.r.t. binary opⁿ multiplication

Above three groups are the only finite groups of real numbers W.r.t. addition and/or Multiplication

Note: If 'e' is the identity element w.r.t. any binary operation '$*$', then $\{e\}$ is always a finite group of order = 1, w.r.t binary opⁿ '$*$'.

Note: In any finite group of order = '2' every element is inverse of itself.



Topic : Finite Group

Cube root of unity are.
$1, \omega, \omega^2$

The set of Cube roots of unity i.e; $\{1, \omega, \omega^2\}$ Form a finite group of order = 3 w.r.t. multiplication

We know $\omega^3 = 1$

1. Closed
2. Associative
3. Identity '1' ∈ Set
4. Inverse

Inverse :- Multiplication Composition table

$(a \cdot b)$	1	ω	ω^2
1	$1=e$	ω	ω^2
ω	ω	ω^2	$\omega^3=1=e$
ω^2	ω^2	$\omega^3=1=e$	$\omega^4=\omega^3\cdot\omega=1\cdot\omega=\omega$

inverse exist for every element

$inv(1) = 1$

$inv(\omega) = \omega^2$

$inv(\omega^2) = \omega$



Topic : Finite Group

Four roots of unity are,
$1, -1, i, -i$

The set of four roots of unity i.e; $\{1, -1, i, -i\}$ Form a finite group of order = 4 w.r.t. multiplication

We Know, $\boxed{i^2 = -1}$

1. Closed ✓
2. Associative ✓
3. Identity $1 \in$ Set ✓
4. Inverse:

$inv(1) = 1$
$inv(-1) = -1$
$inv(i) = -i$
$inv(-i) = i$

inverse exist for every element

Multi-plication $\cdot$	1	-1	i	$-i$
1	$1 = e$	-1	i	$-i$
-1	-1	$1 = e$	$-i$	$+i$
i	i	$-i$	i^2	$-i^2 = -(-1) = 1 = e$
$-i$	$-i$	i	$-i^2 = -(-1) = 1 = e$	i^2

Note: The set of nth roots of unity will form a finite group of order $= n$, w.r.t. multiplication



Topic : Addition modulo 'm' $\oplus_m$

Let 'm' be any fixed positive integer,
for any two non-negative integers a & b,

$$a \oplus_m b = \begin{cases} a+b, & \text{if } (a+b) < m \\ r, & \text{otherwise,} \end{cases}$$

where 'r' is remainder obtained on dividing (a+b) by m.

$\Downarrow$

$(a+b)\,\%\,m$

$2 \oplus_7 3 = 5$

$4 \oplus_7 5 = 2$



Topic : Multiplication modulo 'm' $\otimes_m$

Let 'm' be any fixed positive integer,
for any two non-negative integers, a & b

$$a \otimes_m b = \begin{cases} a \cdot b, & \text{if } (a \cdot b) < m \\ r, & \text{otherwise} \end{cases}$$

$$\Downarrow$$

$$(a \cdot b) \,\%\, m$$

where r is the remainder obtained on dividing $(a \cdot b)$ by m

$2 \otimes_7 3 = 6$

$4 \otimes_7 5 = 6$



Topic : NOTE

$(x + ?) \% n = x$

$(x < n) \cdot 0$

① The set $\{0, 1, 2, 3, \ldots\ldots, (n-1)\}$ form a group w.r.t. binary operation $\oplus_n$

Eg: The set $\{0, 1, 2, 3, 4\}$ form a group of order = 5 w.r.t. binary opⁿ $\oplus_5$

$inv(0) = 0$
$inv(1) = 4$
$inv(2) = 3$
$inv(3) = 2$
$inv(4) = 1$

① Set is closed w.r.t. $\oplus_5$

② $\oplus_n$ is associative for all positive integers 'n'

③ '0' is the identity element w.r.t. $\oplus_n$, for all +ve integer 'n'.

④ '0' is the identity element, $\therefore inv(0) = 0$
for every other element $x \in$ Set, $inv(x) = n - x$ w.r.t $\oplus_n$



Topic : NOTE

② let $A = \{x \mid x \in N \text{ and } x < n \text{ and } GCD(x,n) = 1\}$

it is a set of all natural numbers that are less than 'n' and Co-prime to 'n'

if GCD of two number is '1', then numbers are Co-prime or relatively prime

eg: GCD(4,9) = 1
∴ 4 & 9 are relatively prime (or) Co-prime

let n = 15,

∴ Set = {1, 2, 4, 7, 8, 11, 13, 14}

it is a set of all natural numbers which are less than 15 & Co-prime to 15



Topic : NOTE

③ The set of all natural numbers that are less than 'n' and Co-prime to 'n' form a group w.r.t. binary opⁿ $\otimes_n$

eg let $n = 8$.

then set of all natural no.s that are less than 8 & Co-prime to 8 is $= \{1, 3, 5, 7\}$

$\therefore \{1, 3, 5, 7\}$ form a finite group of order $= 4$

w.r.t. binary operation $\otimes_8$

eg: $\{1, 3, 5, 7\}$ is a group w.r.t. $\otimes_8$

∴ Group

① Set $\{1, 3, 5, 7\}$ is closed w.r.t $\otimes_8$

② $\otimes_n$ is associative for any positive integer 'n'

③ '1' is the identity element w.r.t $\otimes_n$, for any positive integer $n \geq 2$

④ Inverse:

- $1 \otimes_8 1 = (1 \cdot 1) \bmod 8 = 1 = (\text{identity})$
- $3 \otimes_8 3 = (3 \cdot 3) \bmod 8 = 1 = (\text{identity})$
- $5 \otimes_8 5 = (5 \cdot 5) \bmod 8 = 1 = (\text{identity})$
- $7 \otimes_8 7 = (7 \cdot 7) \bmod 8 = 1 = (\text{identity})$

∴ inv(1) = 1

∴ inv(3) = 3

∴ inv(5) = 5

∴ inv(7) = 7

Note:

① $\{1, 3\}$ w.r.t. $\otimes_8$ is a group of order = 2

② $\{1, 5\}$ w.r.t. $\otimes_8$ is a group of order = 2

③ $\{1, 7\}$ w.r.t. $\otimes_8$ is a group of order = 2

④ $\{3, 5\}$ w.r.t. $\otimes_8$ is not a group

⑤ $\{1, 3, 5\}$ w.r.t. $\otimes_8$ is not a group

eg. Let $n = 15$,

∴ Set of elements

s.t. $x \geq 1$ & $GCD(x, 15) = 1$ is $= \{1, 2, 4, 7, 8, 11, 13, 14\}$

∴ $\{1, 2, 4, 7, 8, 11, 13, 14\}$ form a group w.r.t. $\otimes_{15}$

- Find inverse of every element of the above group.

$inv(1) = 1$

$inv(2) = 8$ $\quad$ $inv(8) = 2$

$inv(4) = 4$ $\quad$ $inv(11) = 11$

$inv(7) = 13$ $\quad$ $inv(13) = 7$

$inv(14) = 14$



Topic : NOTE

④ If 'p' is a prime number, then all natural numbers less than 'p' will be co-prime to 'p'.

∴ The set $\{1, 2, 3, \ldots, (p-1)\}$ will form a group w.r.t. binary op^n $\otimes_p$, where 'p' is a prime number.

eg: let $n = 7$ {7 is a prime number}

∴ $\{1, 2, 3, 4, 5, 6\}$ will form a group w.r.t. $\otimes_7$

Q:- Which of the following statement is/are false

(a) False $\{1, 2, 3, 4, 5\}$ form a group w.r.t. $\oplus_6$ — $0 \notin$ set {Not closed}

(b) False $\{1, 2, 3, 4, 5\}$ form a group w.r.t. $\otimes_5$ — $2 \otimes_6 3 = 0 \notin$ set ∴ Not Closed

(c) False $\{0, 1, 2, 3, 4\}$ form a group w.r.t. $\otimes_5$ — inv(0) will not exist.

(d) $\{0, 1, 2, 3, 4\}$ form a group w.r.t. $+_5$

True



Topic : Order of an element in a group (G,*)

Let $(G,*)$ be a group.

for an element $a \in G$, the order of element 'a' is the least positive integer 'n' such that

$$(a)^n = e \text{ (identity element of group } G)$$

$$\Downarrow$$

$$\underbrace{a * a * a * a \ldots\ldots * a}_{n \text{ times } 'a'}$$

order of element 'a' may be denoted by $o(a)$.

eg. We know $\{1, -1\}$ is a finite group of order = 2 w.r.t. binary opn multiplication find order of every element of the group.

$O(G) = 2$

- 1 = identity element

  $(1)^1$ = identity element

  $\therefore$ $\boxed{O(1) = 1}$

- $(-1)^1 = -1 \neq$ identity

  $(-1)^2 = +1$ = identity

  $\therefore$ $\boxed{O(-1) = 2}$

{ Note:- Order of identity element of the group is always = '1' }

eg. We know $\{1, \omega, \omega^2\}$ is a finite group of order = 3 w.r.t. binary opn multiplication find order of every element of the group.

$O(G) = 3$

- 1 = identity $\Rightarrow$ $\therefore$ $\boxed{O(1) = 1}$

- $(\omega)^1 = \omega \neq$ identity

  $(\omega)^2 = \omega^2 \neq$ identity

  $(\omega)^3 = \omega^3 = 1 =$ identity $\Rightarrow$ $\therefore$ $\boxed{O(\omega) = 3}$

$inv(\omega) = \omega^2$

$O(\omega) = O(\omega^2)$

- $(\omega^2)^1 = \omega^2 \neq$ identity

  $(\omega^2)^2 = \omega^4 = \omega^3 \cdot \omega = \omega \neq$ identity

  $(\omega^2)^3 = \omega^6 = \omega^3 \omega^3 = 1 \cdot 1 = 1 =$ identity $\Rightarrow$ $\therefore$ $\boxed{O(\omega^2) = 3}$

eg. We know $\{1, -1, i, -i\}$ is a finite group of order $= 4$ w.r.t. binary opn multiplication find order of every element of the group. $O(G) = 4$

- 1 = identity $\Rightarrow \therefore$ $\boxed{O(1) = 1}$

- $(-1)^1 = -1 \neq$ identity
  $(-1)^2 = 1 =$ identity $\Rightarrow \therefore$ $\boxed{O(-1) = 2}$

- $(i)^1 = i$
  $(i)^2 = -1$
  $(i)^3 = (i)^2 \cdot (i) = -i$
  $(i)^4 = (i)^2 \cdot (i)^2 = -1 \cdot -1 = 1 =$ identity $\Rightarrow \therefore$ $\boxed{O(i) = 4}$

- $(-i)^1 = -i$
  $(-i)^2 = +i^2 = -1$
  $(-i)^3 = (-i)^2 \cdot (-i) = -1 \cdot -i = i$
  $(-i)^4 = (-i)^2 \cdot (-i)^2 = -1 \cdot -1 = 1 =$ identity $\Rightarrow \therefore$ $\boxed{O(-i) = 4}$

$\text{inv}(i) = -i$
&
$O(i) = O(-i)$



Topic : Properties w.r.t. Finite Group

① order of identity element of the group is always = 1

② For finite group 'G', order of an element of group G is less than or equal to the order of group G

i.e. if $a \in G$, then

$$O(a) \leq O(G)$$

③ For an element 'a' belonging to group $(G, *)$,

order of element is always equal to the order of inv(a)

i.e., $O(a) = O(a^{-1})$



Topic : Properties w.r.t. Finite Group

(4) For finite group 'G', order of an element of group G divides the order of group 'G'.

i.e. if $a \in G$, then $O(a)$ divides $O(G)$

(5) Order of an element of the group can be '1' if and only if the element is identity element of the group.

(6) for an element $a \in G$
if $O(a)=2$, then $inv(a)=a$ } i.e. for any element of order = 2, that element is inverse of itself.

eg: We know $\{0, 1, 2, 3, 4\}$ form a group of order = 5. $O(G) = 5$

w.r.t. binary opⁿ $\oplus_5$, then find order of every element of group.

→ $O(G) = 5$ ← it is a prime number

∴ order of every element of the above group must divide '5';

'5' is a prime no. ∴ Possible orders for the elements of the above group are 1 or 5

1 ↓ Possible only for identity element

5 ↓ for every other non-identity element

∴ $O(0) = 1$

$O(1) = 5$

$O(2) = 5$

$O(3) = 5$

$O(4) = 5$

eg: We know $\{0, 1, 2, 3, 4\}$ form a group of order = 5, $O(G) = 5$

w.r.t. binary opⁿ $\oplus_5$, then find order of every element of group.

① 0 = identity

$\therefore$ $O(0) = 1$

② $(1)^1 = 1$

$(1)^2 = 1 \oplus_5 1 = 2$

$(1)^3 = 1 \oplus_5 1 \oplus_5 1 = (1 \oplus_5 1) \oplus_5 1$

$= (1)^2 \oplus_5 1 = 2 \oplus_5 1 = 3$

$(1)^4 = (1)^3 \oplus_5 1 = 3 \oplus_5 1 = 4$

$(1)^5 = (1)^4 \oplus_5 1 = 4 \oplus_5 1 = 0$ = identity

$\therefore O(1) = 5$

$\therefore$ inv(1) = 4

$\therefore O(4) = O(1) \Rightarrow$ i.e. $O(4) = 5$

③ $(2)^1 = 2$

$(2)^2 = 2 \oplus_5 2 = 4$

$(2)^3 = (2)^2 \oplus_5 2 = 4 \oplus_5 2 = 1$

$(2)^4 = (2)^3 \oplus_5 2 = 1 \oplus_5 2 = 3$

$(2)^5 = (2)^4 \oplus_5 2 = 3 \oplus_5 2 = 0$ = identity $\Rightarrow O(2) = 5$

inv(2) = 3

$O(3) = O(2) \Rightarrow$ i.e. $O(3) = 5$



2 mins Summary

Topic Example of groups w.r.t. $\oplus_m$ & $\otimes_m$

Topic Order of an element of the group

Slide

THANK - YOU